

Table 1
Performance Evaluation of Various Prompting Framework in Hadoop

	Audit Trail			Authentication			Heartbeat			Pooling			Scheduling			avg		
	Pre	Re	F1	Pre	Re	F1	Pre	Re	F1	Pre	Re	F1	Pre	Re	F1	Pre	Re	F1
P_{Bas} (D)	0.14	0.50	0.22	0.61	0.76	0.68	0.69	0.75	0.72	0.93	0.46	0.61	0.68	0.75	0.71	0.61	0.64	0.59
P_{ABas} (D)	0.08	0.50	0.14	0.70	0.84	0.76	0.82	0.58	0.68	0.76	0.46	0.57	0.70	0.74	0.72	0.61	0.62	0.57
Prmt4TD (D)	0.20	0.50	0.29	0.70	0.76	0.73	0.82	0.75	0.78	0.93	0.29	0.45	0.76	0.81	0.79	0.68	0.62	0.61
P_{Bas} (Q)	0.25	0.25	0.25	0.66	0.84	0.74	0.72	0.75	0.73	0.93	0.29	0.45	0.78	0.68	0.72	0.67	0.56	0.58
P_{ABas} (Q)	0.12	0.25	0.17	0.65	0.88	0.75	0.83	0.79	0.81	0.91	0.25	0.39	0.71	0.76	0.74	0.64	0.59	0.57
Prmt4TD (Q)	0.50	0.50	0.50	0.77	0.68	0.72	0.75	0.62	0.68	0.95	0.25	0.39	0.80	0.63	0.70	0.75	0.54	0.60
P_{Bas} (G)	0.12	0.25	0.17	0.62	0.84	0.71	0.61	0.58	0.60	0.93	0.33	0.49	0.74	0.66	0.70	0.60	0.53	0.53
P_{ABas} (G)	0.40	0.50	0.44	0.75	0.84	0.79	0.55	0.46	0.50	0.91	0.25	0.39	0.75	0.60	0.67	0.67	0.53	0.56
Prmt4TD (G)	0.33	0.50	0.40	0.78	0.84	0.81	0.65	0.62	0.64	0.96	0.28	0.44	0.81	0.69	0.75	0.71	0.59	0.61

D, Q, and G denote DeepSeek-V3, Qwen3-Max, and Gemini-3.1-Pro, respectively.

Generalizability of Prmt4TD Across Different LLMs:

For the architectural tactics (ATs) detection task, we propose Prmt4TD, a small model-augmented prompting framework. By combining fine-tuned small model with In-Context Learning, it not only reduces the cost of fine-tuning but also provides large language model (LLM) with additional domain knowledge. In the paper, we employed DeepSeek-v3¹, an advanced model known for its outstanding performance and the best value in the market, as the LLM for ATs detection in Prmt4TD.

Essentially, the design of Prmt4TD is model-agnostic. To demonstrate that the framework's performance in the ATs detection task is not a result of model-specific alignment, we conducted additional evaluations by replacing DeepSeek-v3 with two other mainstream LLMs, Qwen3-Max² and Gemini-3.1-Pro³. In addition to applying these LLMs to Prmt4TD, we also applied Qwen3-Max and Gemini-3.1-Pro to the baseline methods P_{Bas} and P_{ABas} under the same experimental conditions, in order to comprehensively evaluate the effectiveness of Prmt4TD. To mitigate any potential bias that the training set (TacticData dataset) might introduce to Prmt4TD, we chose to evaluate the corresponding methods using all the positive samples from the Hadoop case. This is because if the test set (TacticData dataset) were used, Prmt4TD would have an inherent advantage, as its small model component were trained on the same data distribution. To ensure fairness, we chose to conduct the evaluation on the Hadoop case, where all methods start from the same baseline. In the supplementary evaluation, considering practical API cost constraints, we constructed the test set using ATs positive samples. This design allows for an efficient yet focused evaluation while still providing a reliable comparison across models.

The additional experimental results, as shown in Table 1, indicate that regardless of which underlying LLM is used, Prmt4TD (Prmt4TD (D) avg $F1 = 0.61$; Prmt4TD (Q) avg $F1 = 0.60$; Prmt4TD (G) avg $F1 = 0.61$) can effectively

Table 2

A Comparative Study of the Generalization of Prmt4TD

	Prmt4TD (D)			Prmt4TD (Q)			Prmt4TD (G)		
	Pre	Re	F1	Pre	Re	F1	Pre	Re	F1
Audit Trail	0.20	0.50	0.29	0.50	0.50	0.50	0.33	0.50	0.40
Authentication	0.70	0.76	0.73	0.77	0.68	0.72	0.78	0.84	0.81
Heartbeat	0.82	0.75	0.78	0.75	0.62	0.68	0.65	0.62	0.64
Pooling	0.93	0.29	0.45	0.95	0.25	0.39	0.96	0.28	0.44
Scheduling	0.76	0.81	0.79	0.80	0.63	0.70	0.81	0.69	0.75
avg	0.68	0.62	0.61	0.75	0.54	0.60	0.71	0.59	0.61

D, Q, and G denote DeepSeek-V3, Qwen3-Max, and Gemini-3.1-Pro, respectively.

guide the LLM to achieve high accuracy in the ATs detection task compared to the baseline methods. This suggests that the performance improvement brought by Prmt4TD is not due to aligning to any specific capability of DeepSeek-V3, but rather that its core pipeline provides effective domain-specific task knowledge for any LLM with basic reasoning abilities.

We also observe that different LLMs exhibit distinct preferences when applied to Prmt4TD (see Table 2). DeepSeek-V3 demonstrates dominant performance on Heartbeat ($F1 = 0.78$), Pooling ($F1 = 0.45$), and Scheduling ($F1 = 0.79$). Gemini-3.1-Pro achieves the best result on Authentication ($F1 = 0.81$), while Qwen3-Max makes a significant breakthrough on the particularly challenging Audit Trail task ($F1 = 0.50$), substantially outperforming DeepSeek ($F1 = 0.29$). These differences directly reflect variations in pretraining corpora and instruction-tuning alignment across models. However, the overall stability in detection performance provides the evidence of the framework's generalizability.

¹<https://github.com/deepseek-ai/DeepSeek-V3>

²<https://qwen.ai/blog?id=qwen3-max>

³<https://deepmind.google/models/gemini/pro/>